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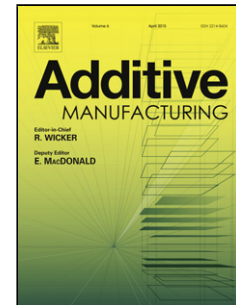
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Finite element simulations of temperature distribution and of densification of a titanium powder during metal laser sintering

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Abstract

Metal Laser Sintering (LS) is a powder bed fusion process that can be used to produce manufactured parts of complex shapes directly from metallic powders. One of the major problems of such powder bed fusion processes is that during the continuous movement of the laser beam, temperature distribution becomes inhomogeneous and instable in the powder. It leads to greater residual stresses in the solidified layer. Thus, temperature analyses must be performed to better understand the heating-cooling process of the powder bed as well as the interactions of different laser scanning paths within a sintering pattern. A transient 3D Finite Element (FE) model of the LS process has been developed with the commercial FE code ABAQUS. The model takes into account the different physical phenomena involved in this powder bed fusion technology (including thermal conduction, radiation and convection). A moving thermal source, modeling the laser scan, is implemented with the user scripting subroutine DFLUX in this FE code. The material's thermal behavior is also defined via the subroutine UMATHT. As the material properties change due to the powder bed fusion process, the model takes it into account. In this way, the calculation of a temperature-dependent behavior is undertaken for the packed powder bed, within its effective thermal conductivity and specific heat. Furthermore, the model accounts for the latent heat due to phase change of the metal powder. Finally, a time- and temperature-dependent formulation for the material's density is also computed, which is then integrated along with the other thermal properties in the heat equation. FE simulations have been applied to the case of titanium powder and show predictions in good agreement with experimental results. The effects of process parameters on the temperature and on the density distribution are also presented.

Keywords: titanium powder, laser sintering (LS)/powder bed fusion, direct metal laser sintering (DMLS), process parameters, finite element method (FEM).

1. Introduction

Additive manufacturing (AM) is driving emerging technologies from the industry that have been developed to create three-dimensional solid parts, of complex shapes, directly from Computer-Aided Design geometries. The common feature of these new technologies is that the final part is built by cumulative deposition of material, layer by layer, with no need for further heavy machining. Such processes made a breakthrough in the prototyping practices since they allow to shorten development time and to decrease the costs [1]. Laser Sintering (LS) that belongs to the powder bed fusion technology is an important branch of AM for potential application such cost-saving practices. It was invented by C.R. Deckard at the University of Texas, U.S.A., in 1986 [2]. In such powder bed fusion technology, a thin layer of powder is spread and is then selectively sintered with a laser beam. This process is repeated until the whole 3D part is fabricated. The powder bed fusion process was initially developed to produce polymer-based components. Nowadays, a wide range of powdered materials can be employed, for further details see the overview of this technology by Kumar [3]. LS of metal powders is also referred as Metal Laser Sintering (MLS). Metal components can be fabricated by different powder bed fusion sub-technologies, such as single- or double-component metal powders processes. The present work is focused on the LS for powder bed fusion process of single metallic powders.

Titanium powders are very interesting since different functional such as aeronautical parts [4] and medical implants parts can be fabricated by LS processes. For example, titanium powders have been used to fabricate dental implants using powder bed fusion processes in Ref. [5]. One advantage of using powder bed fusion technology in this case is that it is possible to fabricate implants with controlled porosity using appropriate processing parameters. Laoui et al. [5] built dental implants with a porous surface layer surrounding a dense core. Their dental implants then possessed a good osseointegration in the body (due to their porous surface) and also good mechanical properties such as tensile and fatigue strength (due to their dense core). As reported in the work of Kruth et al. [6], the basic binding mechanisms of a single metallic powder that can occur in powder bed fusion processes are solid state sintering and/or liquid phase sintering (also called partial melting in Ref. [6]). Moreover both mechanisms can also run simultaneously in agreement with the work of Tolochko et al. [7]. As presented in Ref. [7], the binding mechanisms acting in powder bed fusion processes depend on the laser technology, on components of metallic

powders and most importantly on processing parameters. However Tolochko et al. [7] pointed out that it is difficult to realize this process in practice because of the occurrence of “balling” (formation of large spherical liquid droplets at the surface).

Furthermore, the high intensity of laser beam used in LS processes leads to a rapid heating of a small region of powder layer, which results in great changes of the distribution of temperatures and then large temperature gradients. It causes residual stresses that affect part quality (such as curling defect). Consequently, LS of titanium powder remains a complicated process. In order to prevent such defects and to ensure the achievement of the desired quality of the product, the processing parameters of the powder bed fusion technology must be carefully selected. Classically, the choice of processing parameters can be determined empirically with many tests and failures. This can be replaced by less costly numerical studies, which are more and more feasible, to improve LS processes.

Numerical simulations were previously employed to determine the temperature field under LS of polymer powder bed fusion processes. The commonly used numerical methods for solving LS thermal problems are the Finite Difference (FD) and Finite Element (FE) methods. To save computation time, a one-dimensional numerical model can be developed. For example, Nelson et al. [8] estimated the distribution of temperature and of density by using one-dimensional (1D) FD and FE methods for polycarbonate powders. However 1D models are not always enough to understand the details of thermal and sintering phenomena occurring during powder bed fusion processes. Consequently, two-dimensional (2D) numerical models have been developed both for amorphous [9] and semi-crystalline [10] polymers. In order to better simulate the LS process and to provide accurate information about thermal gradient and densification in the powder bed, three-dimensional (3D) models were developed. Papadatos et al. [11,12] predicted the density and temperature variations in the built parts of amorphous polymer by using a 3D FD method. A 3D FE model for LS of polymer powders was developed by Bugada et al. [13]. In their work, both thermal and sintering phenomena were taken into account by rewriting the thermal and sintering equations for a steady state problem. As for amorphous and semi-crystalline polymer powders, Dong et al. [14-16] developed more recently a 3D transient FE model to predict temperature and density distribution under LS. In the model they presented [14-16], the preheating temperature and different laser process parameters are considered, such as its beam velocity, power, and diameter. Moreover, this model is fully capable of taking into account the powder density and specific heat as a function of temperature, with the thermal conductivity and latent heat also considered here [14-16].

In contrast to polymer powders, only some numerical studies on sintering of metal powders

are available in the literature. As there is no translational or planar symmetry in real sintering processes, Kolossov et al. [17] developed a 3D FE model to predict temperature evolution during the sintering of titanium powders. The temperature dependent thermal conductivity and the specific heat are taken into account in Ref. [17]. However, the density of the material was assumed to be constant in the FE predictions reported there. Xiao and Zhang [18] developed a 3D Finite Volume model of sintering of single-component powder by partial melting. They assumed that melting and resolidification phenomena occur and affect significantly the temperature distribution during sintering. In their work [18], a moving laser beam was considered and a parametric study was carried out using AISI 304 powder. Patil and Yadava [19] developed a 2D FE thermal model to determine transient temperature distributions in a single-component powder layer. The effects of the following process parameters were also investigated [19]: laser power, laser beam diameter, laser on-time, laser off-time and hatch spacing. However, their thermal conductivity of metal powder and their specific heat were assumed to be constant. It should be noted that the hatch spacing is a process parameter specifying the distance between two parallel scanning paths by the laser beam (the scanning being in the two different directions). Recently, Liu et al. [20] proposed a micro-scale approach to analyze the temperature field within the porous powder bed during the sintering of metal powders. Their micro-scale approach is based on a 3D FE thermal analysis of three layers of powder. To study the mechanism of laser sintering, two cases were simulated by Liu et al. [20]: the laser beam struck the last but one top layer of the powder, or the top layer. The temperature variations throughout the powder layer thickness were then predicted. However, predicted temperature distributions were inhomogeneous owing to their discretely distributed particles.

In the present paper, the powder bed fusion process 3D FE model proposed by Dong et al. [14-16] is extended to study the LS of titanium powders. The numerical model is developed with the commercial FE code ABAQUS where was implemented the thermal and sintering sub-models for this metal powder bed fusion process. The simulation is operated by first resolving the density, thermal conductivity, specific heat and latent heat, in accordance with Refs. [14-16]. All these properties are indeed function of temperature, or even temperature and time. Secondly, all the updated values are used as materials properties for solving the heat equation. The material properties and the processing parameters, which can affect the sintering quality, will be further studied to find out the optimal sintering parameters. The results for the titanium powder are presented thereafter in terms of temperature and density distributions. We also compare these predicted results to the experimental ones from the literature [17,19].

2. Modeling approach

The modeling approach developed in the work of Dong and his co-workers [14-16] is briefly reviewed in the following section. In the LS process, the powder bed is pre-heated and we assumed an initial temperature which is lower than the melting temperature of powder material. The laser beam scans the top surface of powder bed following a prescribed direction with a constant velocity. Under the laser beam, heat is transferred to powder material and then a sintering region is formed. After the laser beam moves away, the sintering region cools, thus forming a zone of larger density. Consequently, the numerical model of this powder bed fusion process must include: optical, thermal and sintering sub-models. The optical sub-model refers to the photon-matter interactions, i.e. between the laser and the powder surface (facing the laser beam). The thermal sub-model refers to the heat transfer mechanisms occurring in the material due to the laser penetration within the powder bed, and heat dissipation at surface. The sintering sub-model refers to the sintering process, i.e. the densification phenomenon of the powder towards the fully dense solid material.

2.1. Optical sub-model

LS systems usually use a powerful laser as the thermal energy source for inducing selective sintering. It is known from literature [21] that the distribution of the laser intensity is Gaussian. In accordance with Patil and Yadava [19], the distribution of the laser intensity is taken as:

$$I(r) = \frac{4.55P}{\pi R^2} \exp \left[-4.5 \left(\frac{r}{R} \right)^2 \right] \quad (1)$$

Here, r is the distance from the center of the laser beam, R the known laser beam radius and P the known laser power. Since the laser beam is moving during the powder bed fusion process on this surface, the radial distance r appearing in Eq. (1) also depends on the beam velocity and scanning path. The scanning path thus undertaken through the coordinate system detailed below may then also obey characteristics of the scanning pattern characteristics such as the hatch spacing. Following the work of Dong et al. [15], r is defined in the Cartesian coordinates as:

$$r = \sqrt{(x - v_x t - x_0)^2 + (y - v_y t - y_0)^2} \quad (2)$$

where x and y represent the planar coordinates of the point of the powder bed surface, v_x and v_y the

beam velocity components in x and y directions, respectively. The initial coordinates of the laser beam are represented by x_0 and y_0 .

2.2. Thermal sub-model

During the powder bed fusion process, a powder layer with a very thin thickness is spread on the former powder layers and sintered material. In the sintering region, the absorbed heat is partly conducted into the powder bed and partly lost through radiation and convection on the powder bed surface, see Ref. [22]. The temperature variation inside the current powder layer is governed by the classical heat transfer equation. In a 3D Cartesian domain where the position of a point M is defined by the coordinates (x, y, z) , the heat transfer equation is written as:

$$\rho C_p \frac{\partial T}{\partial t} = \nabla(k \nabla T) + G(x, y, z, t) \quad (3)$$

where T is the temperature, t the time, ρ , C_p , k are the effective density, specific heat and thermal conductivity of the powder bed and G the internal heat source due to laser irradiation here for LS. As a moving laser beam is considered, the internal heat source is defined following the work of Patil and Yadava [19]. The quantity of heat generated by laser irradiation is then expressed following Eq. (1) and Eq. (2) as:

$$G(x, y, z, t) = \frac{I(r)}{s} = \frac{4.55P}{\pi R^2 s} \exp \left(-4.5 \frac{(x - v_x t - x_0)^2 + (y - v_y t - y_0)^2}{R^2} \right) \quad (4)$$

where s is the powder layer thickness.

In the studied powder bed fusion process, the powder bed is assumed pre-heated at a uniform initial temperature T_0 , becoming the initial condition of Eq. (3):

$$T(x, y, z, t = 0) = T_0 \quad (5)$$

In order to solve the heat transfer equation, Eq. (3), the heat fluxes with the thermal surroundings have to be defined. The energy lost through convection and radiation at the powder bed surface is described by the following boundary condition (i.e. at $z = s$, where the coordinate z is

the vertical axis with $z = 0$ at the powder bed bottom):

$$-k \frac{\partial T}{\partial z} \Big|_{z=s} = h(T_0 - T_{z=s}) + \varepsilon_r \sigma (T_0^4 - T_{z=s}^4) \quad (6)$$

where h is the heat convection coefficient, ε_r the surface emissivity of the powder material, σ the Stefan-Boltzmann constant and $T_{z=s}$ the powder bed surface temperature. For surfaces far from the laser scanning area (powder bed walls), they were assumed adiabatic. Consequently, at the bottom surface of the powder bed ($z = 0$) we have no Fourier flux:

$$-k \frac{\partial T}{\partial z} \Big|_{z=0} = 0 \quad (7)$$

It should be noted that, in Eq. (6), the heat convection coefficient h was assumed to be equal to $25 \text{ W/m}^2/\text{K}$ (based on the works of Papadatos [12]) and the surface emissivity of the powder material ε_r taken equal to 0.63, see Ref. [23].

In the present work, the material properties appearing in Eq. (3) were assumed to be temperature-dependent, and the thermal properties of titanium powder are defined by Kolossov et al. [17] as follows. The effective heat conductivity of Titanium powder, k , depends on the temperature as:

$$k = (a + (b - a)\phi) \times k_s \quad (8)$$

where k_s is the heat conductivity of solid titanium, ϕ a sintering potential, a the ratio between the conductivity of loose powder material and the conductivity of the equivalent solid, and b the ratio between the conductivity of the fully sintered material and the one of the equivalent solid. Similar to the work of Kolossov et al. [17], these two ratios (a and b) are assumed to be temperature independent and are taken equal to: $a = 0.01$ and $b = 0.6$. The sintering potential ϕ in Eq. (8) is defined in a point M by:

$$\phi(M, t) = 1 - \exp\left(-\int_0^t \zeta(T(M, s)) ds\right) \quad (9)$$

where t is the time and $\zeta(T)$ a piecewise linear sintered rate defined as:

$$\zeta(T) = \begin{cases} 0 & , T \leq T_s \\ (T - T_s)/180 & , T > T_s \end{cases} \quad (10)$$

where T_s is the sintering temperature of titanium (taken equal to 1,573 K). The calibration parameter 1/180 was chosen so that $\phi = 0.86$ when the titanium powder is held 1 s at 1,933 K, which produced a good fit of the experimental measurements [17]. In Eqs. (3) and (8), the heat conductivity, k_s , and the specific heat, C_p , of solid titanium are temperature-dependent and their values are taken from Ref. [24].

2.3. Sintering sub-model

During the powder bed fusion process, the local bed density evolves since a part of the solid particles of titanium in the powder layer are sintered. The material transformation from powder state to solid state can be described by rate evolution law of the local material density. The rate of change of the titanium powder bed density is described by the following equation [25]:

$$\frac{d\rho}{dt} = (\rho_{\max} - \rho) \times A \times \exp\left(-\frac{E}{RT} - a_x \alpha\right) \quad (11)$$

where ρ is the current density of powder bed, $\rho_{\max}$ the full density of the solid material, R the universal gas constant, E the activation energy, A a pre-exponential factor, a_x a calibration parameter, and α the solid fraction of titanium (over the total solid and liquid titanium). The value $\alpha = 0$ corresponds to liquid phase; whereas $\alpha = 1$ corresponds to solid phase. The value of E for titanium material is roughly taken equal to 560 kJ/mol, see Ref. [26]. The pre-exponential factor A and calibration parameter a_x are assumed to be equal to $5 \times 10^{11} \text{ s}^{-1}$ and 100 respectively. These values were obtained by fitting the calculated density to the experimental results Ref. [27].

In agreement with Ref. [25], it was assumed that melting occurs at a single temperature, T_m . In case the temperature is larger than T_m , the latent heat required for phase change must be considered. For solid to liquid, for instance, it is taken into account by decreasing solid fraction of the material. We used this method, also known as the temperature recovery method, where the

decrease of solid fraction $\Delta\alpha$ is then given by:

$$\Delta\alpha = \frac{C_p}{L} \Delta T \quad (12)$$

Here, L is the latent heat of fusion of titanium ($L = 2.96 \times 10^5$ J/kg [19]). In Eq. (11), the parameter α represents the solid fraction of titanium in the powder bed. This parameter varies from 1 to 0 respectively before and after the melting temperature T_m , see Ref. [25]. It varies from 0 to 1 at the resolidification. During the heating phase in the FE simulations, when there is no latent heat, a temperature rise ΔT is calculated by the FE code within the time increment Δt . Once the material temperature reaches the melting temperature T_m and if the heat source continues to heat up the material, the temperature will however remain constant for a short time period. During this time, the solid fraction begins to decrease until all of the material is melted (i.e. $\alpha = 0$). Then, at $\alpha = 0$, the temperature restarts to increase from the melting temperature.

3. Procedure for FE analysis

Combining the sub-models presented above, the temperature field in the powder bed is obtained using the commercial FE code ABAQUS/Standard. For heat transfer problems, ABAQUS/Standard uses the Newton method to solve the nonlinear system obtained from the weak form of energy balance equation [28]. Since the material properties are temperature dependent, see Eq. (8) to Eq. (12), they are first updated at each time increment and then used as material constants to update the Jacobian matrix via a user-subroutine UMATHT [28]. In the powder bed fusion process simulations, the heat generated by the laser (determined with Eq. (4)) is implemented via a user-subroutine DFLUX [28]. The scanning time, hatch spacing and laser beam velocity are also implemented within the subroutine. Further details can be found in the work of Dong and co-workers [14-16].

The FE model developed for powder bed fusion process simulations of titanium powder is presented in Fig. 1. The powder bed is assumed as a parallelepiped specimen with the dimensions of 2 mm×1 mm×0.5 mm. The specimen is meshed with solid (continuum) elements called DC3D8 in the ABAQUS element library [28]. It is an 8-node linear brick element with a single degree of freedom (temperature) at each node. In order to ensure the computational precision, the sintered and adjacent zones are discretized with a fine mesh. For this whole geometrical approximation, the 3D

FE model is meshed by 45,248 nodes and 40,500 elements.

4. Results and discussion

4.1. Validation of FE model

In order to check the accuracy of the proposed FE model, preliminary calculations were performed and compared with the experimental results of Kolossov et al. [17]. Thus, the laser path follows the same path defined in the experiments presented in the work of Kolossov et al. [17]. The laser path is defined as function of coordinates and also of laser velocity, Eq. (2). It should be noted that the laser beam scanning path is defined in the y -direction; hence $v_x = 0$ and $v_y = v$. The laser path is marked by the dashed line in Fig. 1. The coordinates followed by the laser on the powder bed surface are plotted in Fig. 2. The material properties, laser parameters and experimental conditions used by Kolossov et al. [17] are summarized in Table 1.

The predicted temperature distribution is presented in Fig. 3. This prediction was obtained just when the laser beam achieved to scan the path pattern represented in Fig. 2. In accordance with [17] and Fig. 2, the laser followed a first straight line along y -direction and then for its second, last, straight line stops at the level of the middle point of the first straight line. In Fig. 3, the highest value of temperature is reached under the laser beam center and it rises to 2,741 K, which is close to the experimental value of $2,673 \pm 200$ K [17]. Moreover, the predicted transverse and longitudinal temperature distributions are plotted respectively in Fig. 4 and Fig. 5 and are clearly in fairly good agreement with the experimental results of Kolossov et al. [17]. Fig. 4 presents the temperature profile along x -direction at the coordinate of $y = 2.5$ mm. In this figure, the maximum value of temperature (2,741 K) is reached near the center of the laser beam. This figure also shows a gradual decrease of the temperature with the distance from the laser center. However, it should be noted that the temperature profile is not symmetric in Fig. 4 since a steep decrease of temperature occurs in the right side of the powder bed surface. With the effect of the previous laser trace and the small hatch spacing, the temperature has a higher value in the left side of the powder bed surface than the other side. Fig. 5 illustrates the temperature distribution along y -direction at the coordinate of $x = 1.85$ mm. As expected, the maximum value of temperature is predicted at laser center. The computed temperature distribution is symmetric because the diffusion heat of the previous laser trace affects the temperature on either side of the laser center in this track. It is interpreted that there is no subtle heat diffusion effect along the y -axis due to the high laser scan speed. At the two ends of the simulated temperature curve, the inflexions represent the boundaries of the laser path.

4.2. Influence of process parameters

Several independent process parameters can influence the sintering process. Such parameters are: laser power, scanning speed, laser beam radius, hatch spacing and pre-heating temperature of the powder bed. The effects of these key parameters on temperature and density distributions of titanium powder material are investigated by FE simulations and are discussed in the following sections. During the FE analysis of each process parameter, one parameter is studied and the others are kept constant to ensure that the effect of the studied parameter is well captured within this sensitivity analysis. The parameters of this sensitive analysis were chosen to mostly reflect the different kinds of influence on the AM part density and temperature profile, using the process parameters of the aforementioned validating experiment as a reference. However, for applications purpose, the process parameter range should be investigated within the specifications of the used powder bed fusion machine, also with the AM parts application dictating quality criteria to be achieved, which in itself is out of the scope of the current study. The choice of a whole set of parameters should also take into account the final optimal density, excessive heating phenomena, as well as other process considerations such as the used powder recycling strategy that is strongly linked to the “pre-heating” temperature presented in this section. An optimization procedure is therefore needed to define the optimal process parameters based on optimal final properties and dimensions of the built part.

a) Laser power

As it is well established in the literature, the laser power has a significant effect on the powder bed surface temperature and on the relative density after sintering. FE simulations are carried out for different values of laser power: 2 W, 3 W and 4 W. The other process parameters and material properties are kept at the values listed in Table 1. The temperature distribution of the powder bed surface (along y-axis at $x = 1.85$ mm) is presented in Fig. 6. The temperature distribution is plotted for the three different values of laser power. As expected, the peak value increases with laser power, see Fig. 6. The same phenomenon of temperature increase was observed along the x-axis in Fig. 7 (at $y = 2.5$ mm). Moreover, the heated area increases since more heat flux is transmitted to the powder bed when laser power increases. The increase of temperature is more obvious in Fig. 7. These results have the same trends as numerical results reported by other authors, for instance Patil and Yadava [19]. In agreement with the work Ref. [27], the sintered depth of

titanium powder by laser was assumed to be close to 100 μm . Consequently, in our FE simulations, the density of the powder bed is taken when it is stable and for a reference depth chosen equal to 100 μm . The predicted relative density of the powder bed, after sintering, is plotted as a function of laser power in Fig. 8. The relative density of the powder bed increases almost linearly with the increase of laser power, see Fig. 8. It should be noted that the effect of hatch spacing was not taken into account. With the parameters chosen for FE simulations, a laser power value of 4 W seems to result in the best relative density (approaching 100%). However, the temperature value obtained here is over the boiling point of titanium under equilibrium conditions for the small concerned volume. This phase change is not modeled in the current approach, and the power from 2 to 4 W should then be carefully considered for the greater laser powers due to this reason. However, this power range is illustrated here for the sake of exploring and extrapolating the performances for laser powers larger than 2 W.

b) Scanning speed

In their numerical study of polymer powder bed fusion process, Dong et al. [15] reported that the temperature of the powder bed surface is significantly affected by scanning speed. Simulations were also performed in this work to investigate this parameter with different scanning speed values, whereas the other process parameter values were kept constant. The chosen values of scanning speed are: 0.5 mm/s, 1 mm/s, and 1.5 mm/s. The simulated temperature distributions of the powder bed surface are presented in Fig. 9. The prediction along y -axis at a position $x = 1.85$ mm is shown in Fig. 9a, while the prediction along x -axis at a position $y = 2.5$ mm is shown in Fig. 9b. We can see that the increase of the laser scanning speed lead to the decrease of the temperature, see Fig. 9. When the scanning speed increases, the laser beam spends less time on the powder bed surface. Therefore, the heat energy from the laser transmitted to the powder bed surface decreases and the amount of non-sintered powder particles tends to increase. Fig. 10 illustrates that the predicted densification of the powder bed decreases with increasing scanning speed. As expected, scanning speed has a significant effect on density. However, from the speed of 1 mm/s, the density and then the energy absorbed by the material is almost stable. The predicted densification of the powder bed follows the experimental trend presented in the work of Song et al. [29] on the LS melting of a titanium powder.

c) *Laser radius*

As it is well known, the laser energy density applied on the powder bed surface is function of laser power, laser scanning speed and laser beam diameter (or radius). The effects of the two first process parameters were studied above. Hence, we decided to study the influence of laser radius on temperature distributions and also on the densification of the powder bed. Three different laser radius values (0.02 mm, 0.025 mm, and 0.05 mm) are used in the FE simulations of the investigated powder bed fusion process. The simulated temperature distributions of the powder bed surface are presented in Fig. 11. The prediction along y-axis at a position $x = 1.85$ mm is shown in Fig. 11a, while the prediction along x-axis at a position $y = 2.5$ mm is shown in Fig. 11b. It should be reminded that the laser scanned one and a half straight line on the powder bed as shown in Fig. 2. With the same laser power $P = 2$ W, scanning speed $v = 1$ mm/s and hatch spacing of 0.1 mm, the increase of the laser radius leads to a decrease of the surface temperature in this particular case, see Fig. 11. This decrease is difficult to envision from the used equations only. It is rather linked to several processing and physical parameters involved in the simulated AM process. In Fig. 11b, it can be noticed that the area sintered by the laser slightly decreases with the increase of laser radius. The FE prediction presented in Fig. 12 illustrates that the increase of laser radius should lead to a non-linear decrease of the relative density. It should be noted that the same trend is also obtained with the effect of laser scanning speed.

d) *Hatch spacing*

Hatch Spacing (HS) is the distance between two neighboring scan lines on the powder bed surface. HS is usually less than the laser beam diameter. In the LS FE simulations performed, the powder bed fusion process parameters are also kept at the values listed in Table 1 with the laser power $P = 2$ W. Three different HS (0.01 mm, 0.015 mm, and 0.02 mm) are used in this study. Fig. 13 presents the effect of the HS on temperature distribution along the y-direction, for an x-coordinate located in the middle of the two neighboring laser paths. The temperature distributions plotted in Fig. 13 show that the temperature decreases with the increase of HS. However, the influence of hatch spacing on the predicted temperature distributions is weak since the chosen values of hatch spacing are close. The effect of hatch spacing on the predicted relative density is shown in Fig. 14. The predicted relative density decreases slightly with the increase of hatch spacing. Nevertheless, the value of hatch spacing must be chosen carefully. Longer hatch spacing leads to discontinuous densification of the powder, and very short hatch spacing will result in over-

sintering.

e) Pre-heating temperature

The last process parameter studied in the work is the pre-heating temperature which is perhaps the most important process parameter in the powder bed fusion process. It should also be noted following Eq. (6) that the “pre-heating/initial” temperature is also considered in the heat dissipation phenomena during the whole process time, i.e. it is the sink temperature for the radiation of the surface as well as the ambient air temperature for convection. Such powder bed fusion process parameter has a significant influence on the LS-processed parts such as: relative density, dimension precision, mechanical properties, etc. It is also to be considered with recycling strategies of the powder that may be subjected to aging in the process.

The FE simulations were carried out for extremely different pre-heating temperatures: 293 K, 750 K and 1,500 K. The two extreme values correspond to room temperature and to a temperature close to the melting temperature of titanium. The other process parameters are kept the same as in Table 1, with the laser power $P = 2$ W. Fig. 15 shows the FE predictions of temperature distribution of the powder bed surface along y -axis at a coordinate $x = 1.85$ mm. As expected, the temperature profile increases with the pre-heating temperature. Under laser irradiance, with the pre-heating temperature of 1,500 K, the temperature in our design of the powder bed is everywhere above the sintering temperature of titanium, see $T_s = 1,573$ K in Eq. (10). Fig. 16 shows the predicted temperature distribution along x -axis at a coordinate $y = 2.5$ mm. The numerical trend presented in Fig. 15 is also naturally found in Fig. 16. Finally, the effect of this pre-heating/sink temperature on the maximum relative density is plotted in Fig. 17. As expected, the predicted relative density increases with the pre-heating temperature.

5. Conclusion

In this paper, 3D finite element simulations of the laser sintering for a titanium powder bed fusion process were performed using the software package ABAQUS. The heat source generated by the laser beam, following a Gaussian intensity distribution, was introduced in ABAQUS via a scripting user-subroutine DFLUX. Furthermore, the heat equation and sub-models describing and the powder sintering and the effect of latent heat were coupled with ABAQUS via its user-subroutine UMATHT.

The predicted temperature distribution over the surface of powder layer was compared to the

experimental results of Kolossov et al. [17]. Reasonable agreements were found between the predicted temperature distributions and the experimental ones. The effects of different processing parameters have been then investigated such as laser power, scanning speed, laser beam radius, hatch spacing and pre-heating temperature. The intensity of heat energy is of the essence as for the influence of these process parameters. A higher temperature distribution, and higher relative density, can be achieved by higher laser power and pre-heating temperature, and/or by lower scanning speed, laser radius and hatch spacing. Generally, non-linear relationships on the density or temperature were found due to the nonlinear nature of the modeled heat transfer, as well as the phase change effects occurring during the process of sintering.

With our numerical model, the process parameters can be accurately chosen to avoid known weaknesses of this additive manufacturing technology, such as the excessive or insufficient heating. However, the first focus of this paper is to propose the 3D-FEM model and to validate it by comparison to existing experimental results. The second focus is to show that the model can predict the influence of the different process parameters. The comparison to experimental results is limited to laboratory experiments only. Next step will be to simulate real AM of 3D objects and use the simulations for process parameters optimization. This optimization can be conducted by setting an objective function that describes the required final properties of the built object within a certain range. In addition, this objective function should also include the dimensional tolerance by imposing the limited rang for Z-bonus as well as the bonuses in the x and y directions [11]. The model will then be a part of an iterative optimization simulation where the process parameters will be changed (within a range) until the set objectives are met.

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References

- [1] X. Yan, P. Gu, A review of rapid prototyping technologies and systems, *Comput. -Aided Des.* 28 (1996) 307-318. DOI: 10.1016/0010-4485(95)00035-6.
- [2] C.R. Deckard, Part generation by layerwise selective sintering, M. Sc. thesis, University of Texas, Austin, USA (1986).
- [3] S. Kumar, Selective laser sintering: A qualitative and objective approach, *JOM* 55 (2003) 43-47. DOI: 10.1007/s11837-003-0175-y.
- [4] S. Das, J.J. Beaman, M. Wohler, D.L. Bourell, Direct laser freeform fabrication of high performance metal components, *Rapid Prototyp. J.* 4 (1998) 112-117. DOI: 10.1108/13552549810222939.
- [5] T. Laoui, E. Santos, K. Osakada, M. Shiomi, M. Morita, S.K. Shaik, N.K. Tolochko, F. Abe, M. Takahashi, Properties of titanium dental implant models made by laser processing, *Proc. Inst. Mech. Eng. , Part C* 220 (2006) 857-863. DOI: 10.1243/09544062JMES133.
- [6] J.-. Kruth, G. Levy, F. Klocke, T.H.C. Childs, Consolidation phenomena in laser and powder-bed based layered manufacturing, *CIRP Ann. Manuf. Technol.* 56 (2007) 730-759. DOI: 10.1016/j.cirp.2007.10.004.
- [7] N.K. Tolochko, M.K. Arshinov, A.V. Gusarov, V.I. Titov, T. Laoui, L. Froyen, Mechanisms of selective laser sintering and heat transfer in Ti powder, *Rapid Prototyp. J.* 9 (2003) 314-326. DOI: 10.1108/13552540310502211.
- [8] J.C. Nelson, Selective laser sintering: a definition of the process and an empirical sintering model, Ph. D. thesis, University of Texas, Austin, USA (1993).
- [9] T.H.C. Childs, M. Berzins, G.R. Ryder, A. Tontowi, Selective laser sintering of an amorphous polymer—simulations and experiments, *Proc. Inst. Mech. Eng. , Part B* 213 (1999) 333-349. DOI: 10.1243/0954405991516822.
- [10] A.E. Tontowi, T.H.C. Childs, Density prediction of crystalline polymer sintered parts at various powder bed temperatures, *Rapid Prototyp. J.* 7 (2001) 180-184. DOI: 10.1108/13552540110395637.
- [11] A.L. Papadatos, S. Ahzi, C.R. Deckard, F.W. Paul, On dimensional stabilities: Modeling of the Bonus-Z during the SLS Process, 8th Solid Freeform Fabr. (SFF) Symp. Proc. (1997) 709-715. URL: <http://sffsymposium.engr.utexas.edu/1997TOC>.
- [12] A.L. Papadatos, Computer simulation and dynamic control of the selective laser sintering process, M. Sc. thesis, Clemson University, Clemson, USA (1998).
- [13] G. Bugeda, M. Cervera, G. Lombera, Numerical prediction of temperature and density distributions in selective laser sintering processes, *Rapid Prototyp. J.* 5 (1999) 21-26. DOI: 10.1108/13552549910251846.
- [14] L. Dong, A. Makradi, S. Ahzi, Y. Rémond, Finite element analysis of temperature and density distributions in selective laser sintering process, *Mater. Sci. Forum* 553 (2007) 75-80. DOI: 10.4028/www.scientific.net/MSF.553.75.

- [15] L. Dong, A. Makradi, S. Ahzi, Y. Remond, Three-dimensional transient finite element analysis of the selective laser sintering process, *J. Mater. Process. Technol.* 209 (2009) 700-706. DOI: 10.1016/j.jmatprotec.2008.02.040.
- [16] L. Dong, A. Makradi, S. Ahzi, Y. Remond, X. Sun, Simulation of the densification of semicrystalline polymer powders during the selective laser sintering process: Application to Nylon 12, *Polym. Sci. , Ser. A* 50 (2008) 704-709. DOI: 10.1134/S0965545X0806014X.
- [17] S. Kolossov, E. Boillat, R. Glardon, P. Fischer, M. Locher, 3D FE simulation for temperature evolution in the selective laser sintering process, *Int. J. Mach. Tools Manuf.* 44 (2004) 117-123. DOI: 10.1016/j.ijmachtools.2003.10.019.
- [18] B. Xiao, Y. Zhang, Partial melting and resolidification of metal powder in selective laser sintering, *J. Thermophys. Heat Transfer* 20 (2006) 439-448. DOI: 10.2514/1.17904.
- [19] R.B. Patil, V. Yadava, Finite element analysis of temperature distribution in single metallic powder layer during metal laser sintering, *Int. J. Mach. Tools Manuf.* 47 (2007) 1069-1080. DOI: 10.1016/j.ijmachtools.2006.09.025.
- [20] F.R. Liu, Q. Zhang, W.P. Zhou, J.J. Zhao, J.M. Chen, Micro scale 3D FEM simulation on thermal evolution within the porous structure in selective laser sintering, *J. Mater. Process. Technol.* 212 (2012) 2058-2065. DOI: 10.1016/j.jmatprotec.2012.05.010.
- [21] P.F. Jacobs, *Rapid prototyping & manufacturing: fundamentals of stereolithography*, Society of Manufacturing Engineers, One SME Drive, PO Box 930, Dearborn, MI 48121-0930, USA, 1992.
- [22] M.M. Sun, J.J. Beaman, A three dimensional model for selective laser sintering, 2nd Solid Freeform Fabr. (SFF) Symp. Proc. (1991) 102-109. URL: <http://sffsymposium.engr.utexas.edu/1991TOC>.
- [23] MatWeb LLC, MatWeb: Online materials information resource (accessed 05.26.2015).
- [24] Y.S. Touloukian, C.Y. Ho, P.E. Liley, *Thermophysical properties of matter, The TPRC Data series*, IFI/Plenum Data Corporation, New York, USA, 1970.
- [25] J.D. Munoz, A. Arizmendi, A. Mendoza-Allende, J.A. Montemayor-Aldrete, High temperature activation energy for plastic deformation of titanium carbide single crystals as a function of the C : Ti atom ratio, *J. Mater. Sci.* 32 (1997) 3189-3193. DOI: 10.1023/A:1018654818631.
- [26] T.H.C. Childs, A.E. Tontowi, Selective laser sintering of a crystalline and a glass-filled crystalline polymer: Experiments and simulations, *Proc. Inst. Mech. Eng. , Part B* 215 (2001) 1481-1495. DOI: 10.1243/0954405011519330.
- [27] P. Fischer, N. Karapatis, V. Romano, R. Glardon, H.P. Weber, A model for the interaction of near-infrared laser pulses with metal powders in selective laser sintering, *Appl. Phys. A: Mater. Sci. Process.* 74 (2002) 467-474. DOI: 10.1007/s003390101139.
- [28] ABAQUS, *Manuals version 6.5*, Dassault Systèmes, Simulia Corp., Providence, RI, USA (2005).
- [29] B. Song, S. Dong, H. Liao, C. Coddet, Process parameter selection for selective laser melting of Ti6Al4V based on temperature distribution simulation and experimental sintering, *Int. J. Adv. Manuf. Technol.* 61 (2012) 967-974. DOI: 10.1007/s00170-011-3776-6.

[30] L. Dong, N. Barth, J. P. M. Correia, S. Ahzi, Modeling and numerical simulation of selective laser sintering, 17th Int. Conf. Therm. Mech. Simul. Exp. Microelectron. Microsyst. (EuroSimE). IEEE Proceedings (2016) 1-4. DOI: 10.1109/EuroSimE.2016.7463376.

Figures Caption list

Figure 1: FE mesh used in the simulations of the powder bed fusion process. z-axis is the vertical, perpendicular to the powder bed surface in the front side that is also the latest powder layer. The studied scanning path is highlighted by the arrows (dashed line the first path and solid line for the half-distance return).

Figure 2: Laser path and relative coordinates used thereafter for the powder bed surface (dashed line the first path and solid line for the half-distance return). The hatch space is the distance between these two lines (0.1 mm here).

Figure 3: Temperature distribution predicted for a laser power $P = 2$ W moving with a velocity $v = 1$ mm/s. The model is in the same configuration as Fig. 1.

Figure 4: Comparison between predicted (solid line) and experimental (dash line) temperature profiles along x-axis, with $y = 2.5$ mm.

Figure 5: Comparison between predicted (solid line) and experimental (dash line) temperature profiles along y-axis, with $x = 1.85$ mm.

Figure 6: Temperature distribution along y-axis predicted by FE simulations for different values of laser power, with $x = 1.85$ mm [30].

Figure 7: Temperature distribution along x-axis predicted by FE simulations for different values of laser power, at $y = 2.5$ mm.

Figure 8: Relative density predicted by FE simulations as a function of laser power at the reference depth of 100 μm . The initial density of the powder is at the bottom of the relative density axis (52.3%). Solid titanium is at 100% (i.e. fully sintered/melted-resolidified titanium powder).

Figure 9: Influence of scanning speed on the predicted temperature distributions along (a) y-axis at $x = 1.85$ mm; and (b) x-axis at $y = 2.5$ mm.

Figure 10: Relative density predicted by FE simulations as a function of scanning speed (at the reference depth of 100 μm). The initial density of the powder is at the bottom of the relative density axis (52.3%).

Figure 11: Influence of laser beam radius on the predicted temperature distributions along (a) y -axis at a position $x = 1.85$ mm and (b) x -axis at $y = 2.5$ mm.

Figure 12: Relative density predicted by FE simulations as a function of laser beam radius (at the reference depth of 100 μm). The initial density of the powder is at the bottom of the relative density axis (52.3%).

Figure 13: Influence of hatch spacing on the predicted temperature distributions along y -axis, x -coordinate located in the middle of the two neighboring laser paths.

Figure 14: Relative density predicted by FE simulations as a function of hatch spacing (at the reference depth of 100 μm). The initial density of the powder is at the bottom of the relative density axis (52.3%).

Figure 15: Temperature distribution along y -axis predicted by FE simulations for different values of pre-heating temperature, at the coordinate $x = 1.85$ mm.

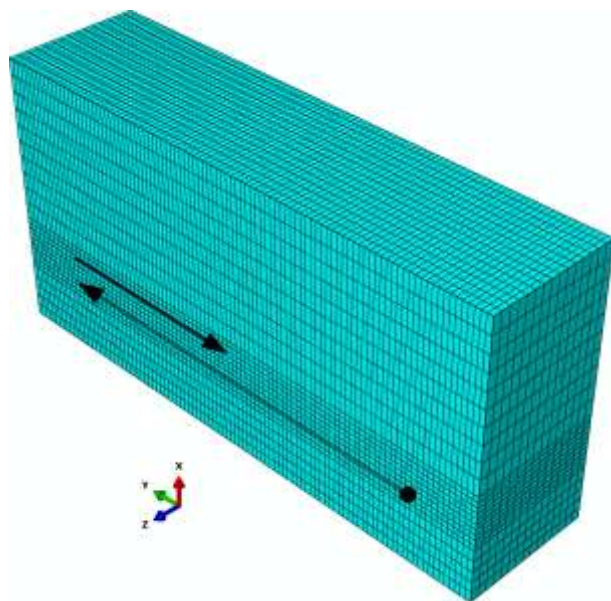
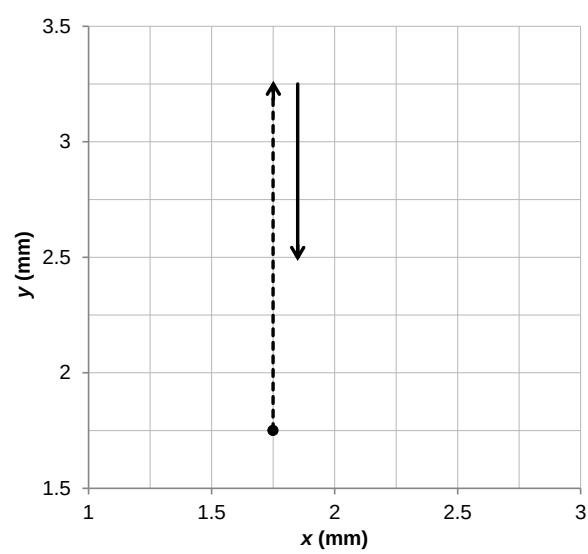
Figure 16: Temperature distribution along x -axis predicted by FE simulations for different values of pre-heating temperature, at the coordinate $y = 2.5$ mm.

Figure 17: Relative density predicted by FE simulations as a function of pre-heating temperature (at the reference depth of 100 μm). The initial density of the powder is at the bottom of the relative density axis (52.3%).

Table list:

Table 1: Titanium material properties and laser parameters used in the experiments of Kolossov et al. [17] and in the FE simulations.

Titanium material properties	
Activation energy (kJ/mol)	560
Surface emissivity of the material ε_r	0.6
Initial density ρ_0 (kg/m ³)	2931.5
Full density of solid material ρ_{max} (kg/m ³)	5605.2
Latent heat (kJ/kg)	296
Melting temperature (K)	1933
Laser parameters and initial conditions	
Initial bed temperature (K)	293
Thickness of powder bed (mm)	0.5
Laser average power (W)	2
Track length (mm)	1.5
Hatch spacing (mm)	0.1
Laser beam radius (mm)	0.025
Scanning speed (mm/s)	1

Figures preview list**Fig. 1****Fig. 2**

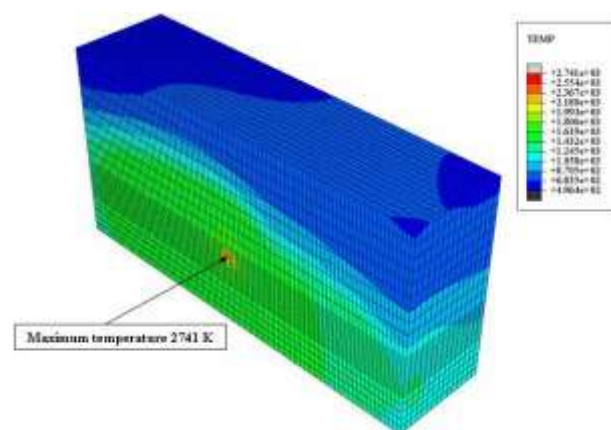


Fig. 3

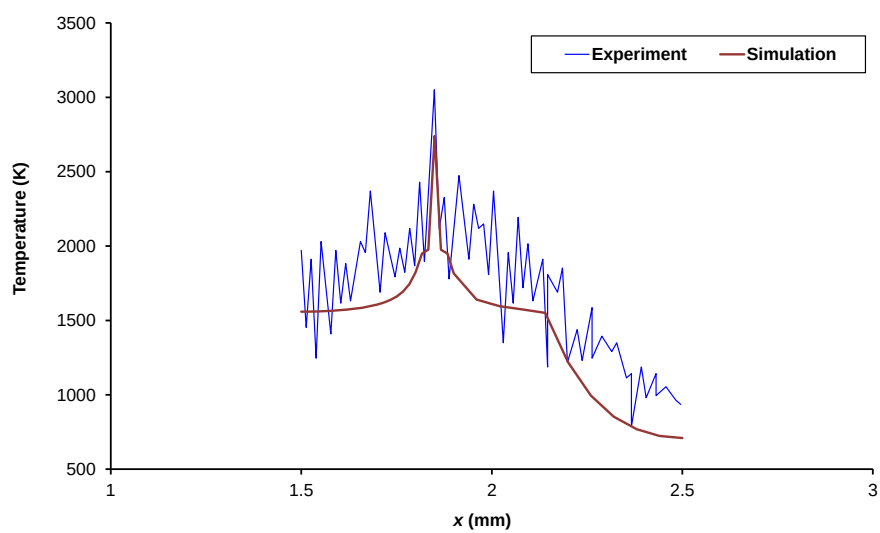
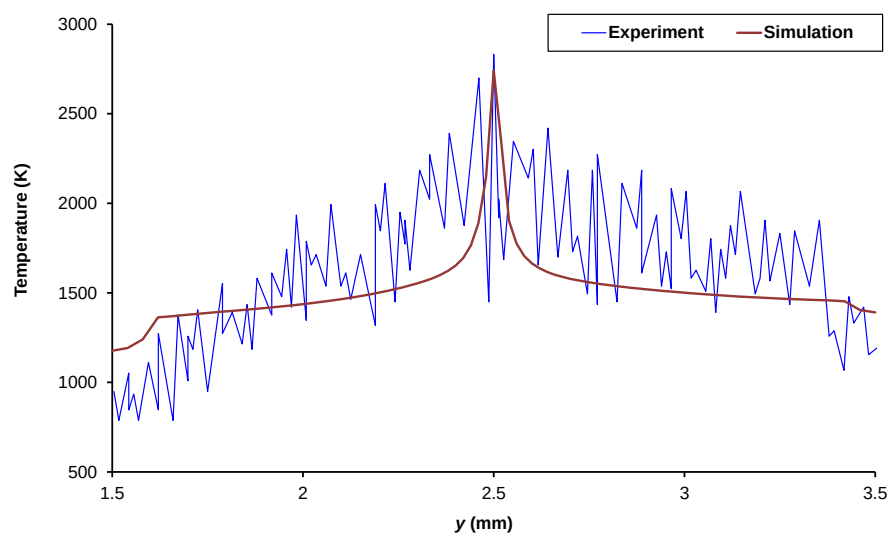
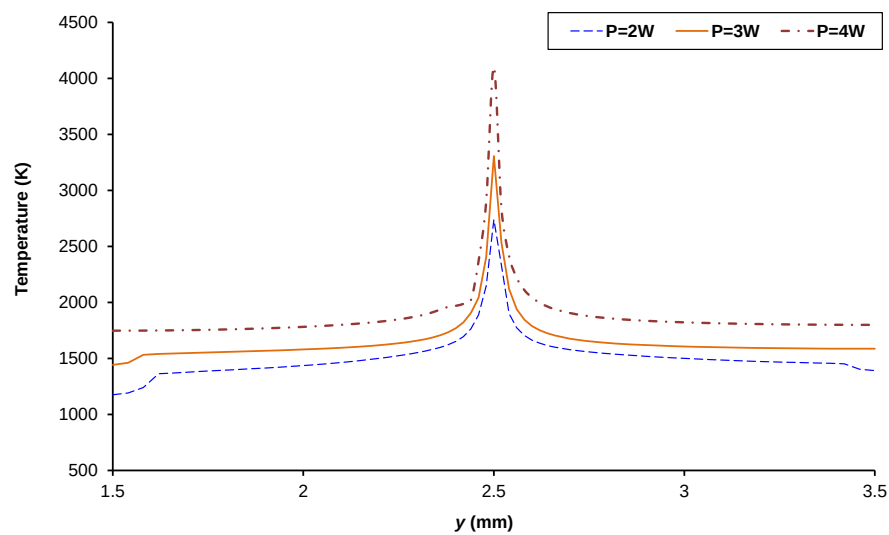
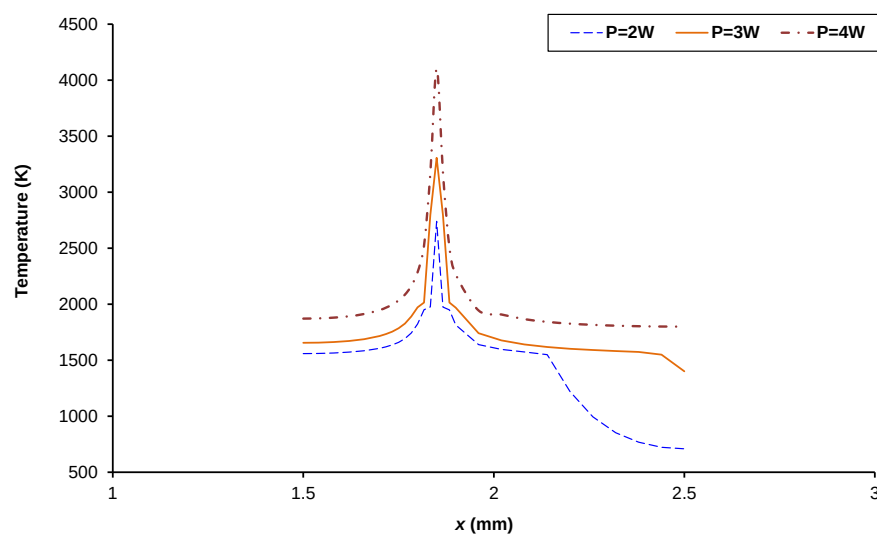
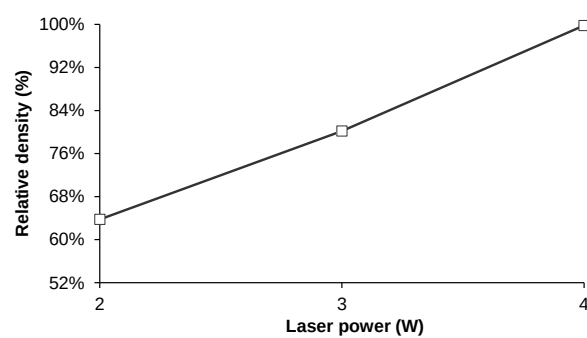
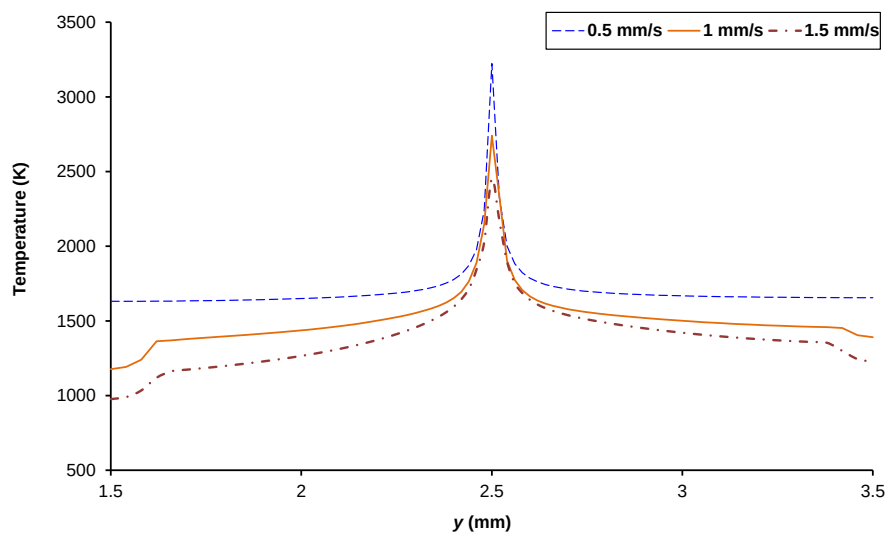


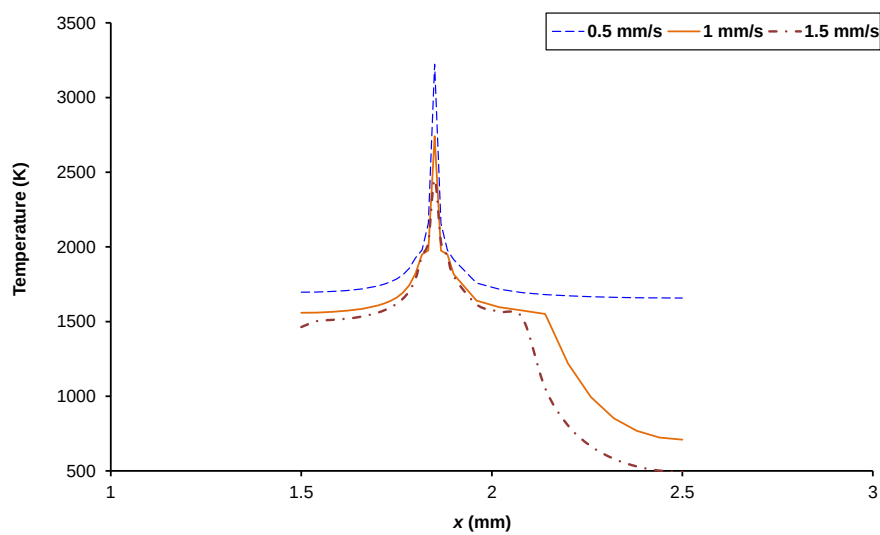
Fig. 4

**Fig. 5****Fig. 6**

**Fig. 7****Fig. 8**

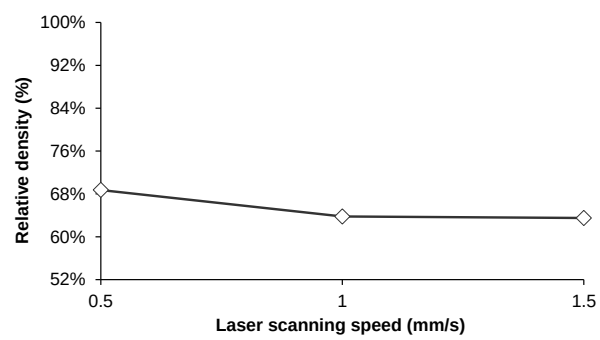


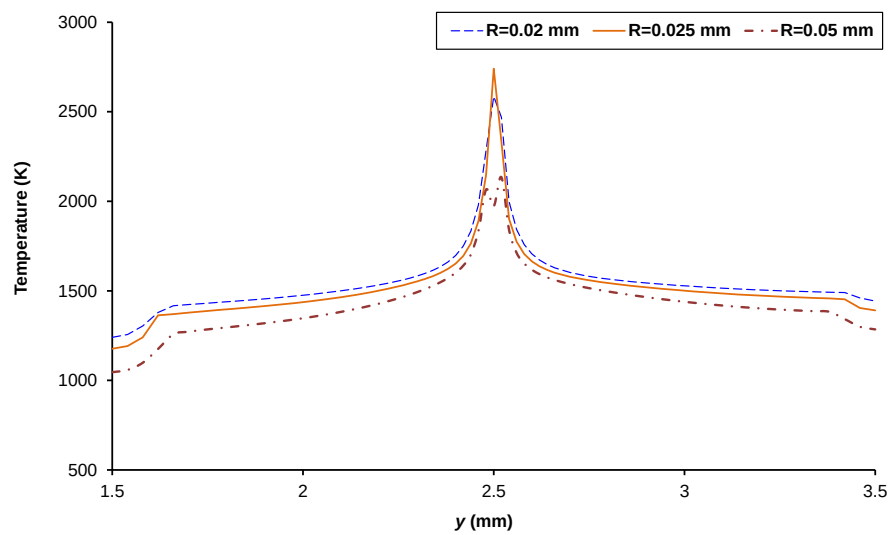
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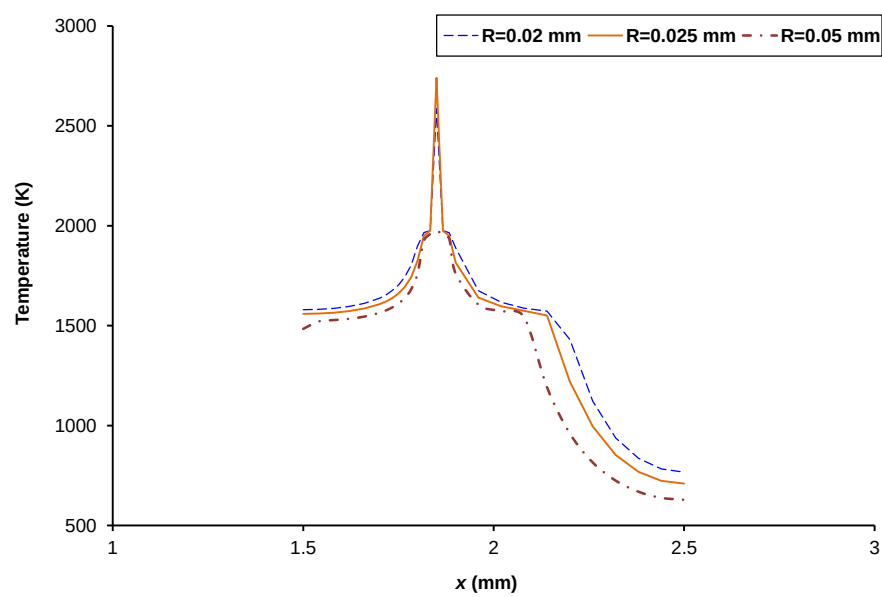
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Fig. 9

**Fig. 10**

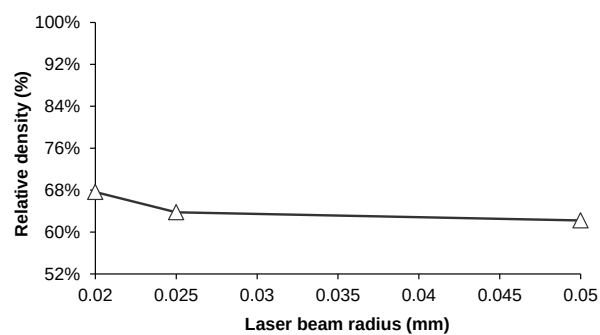
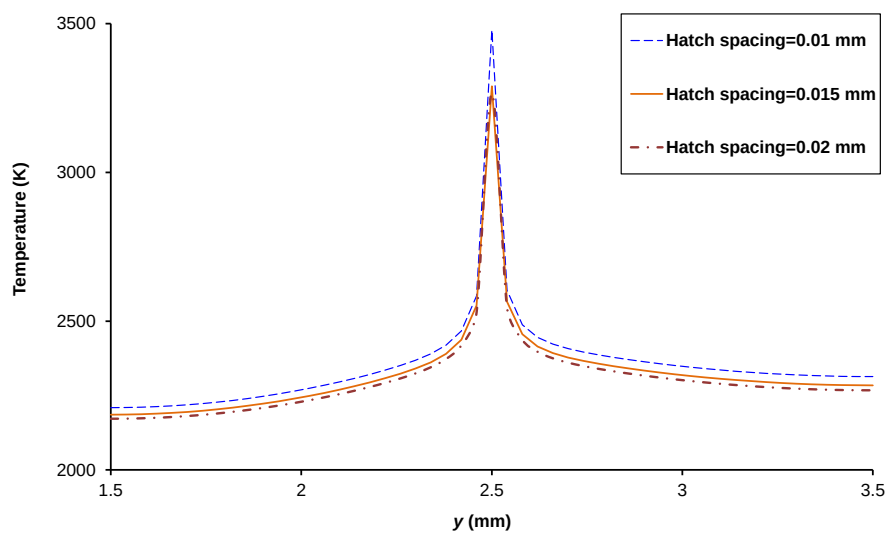


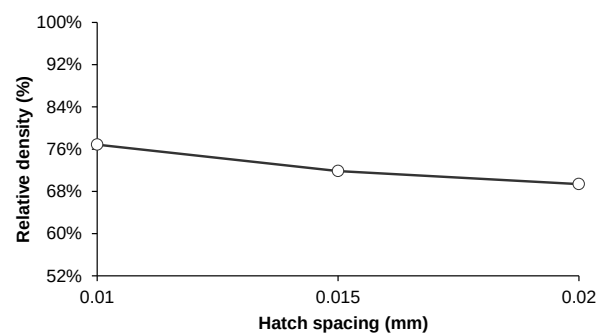
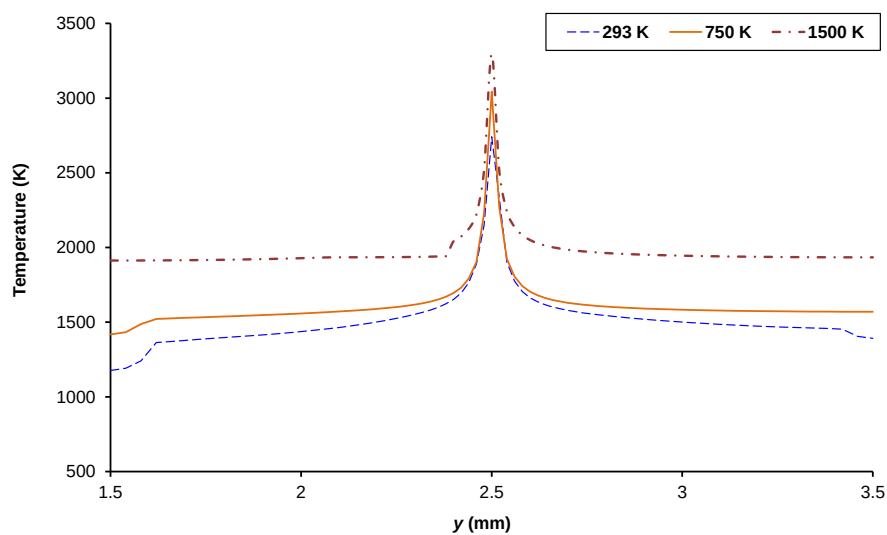
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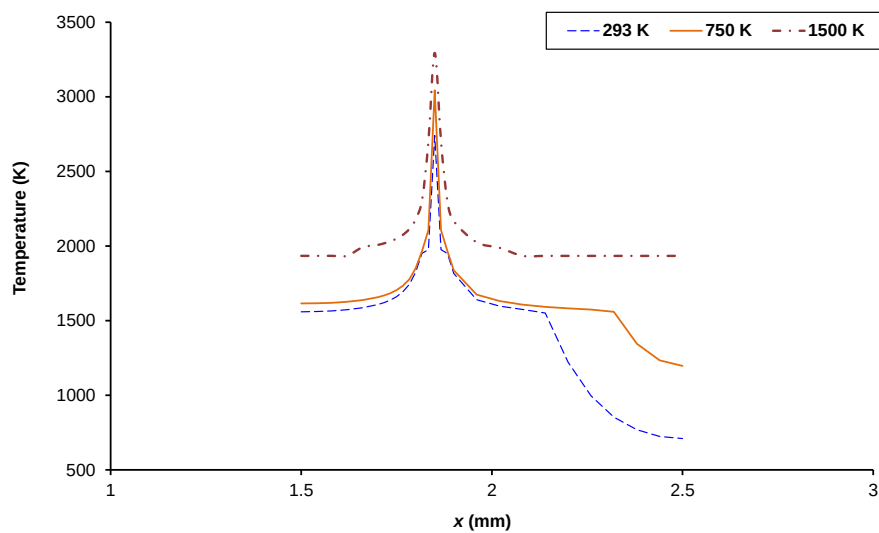
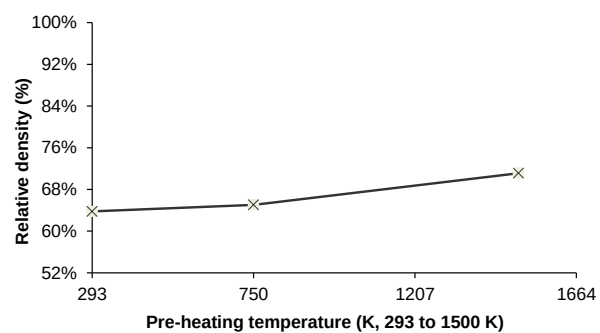


(b)

Fig. 11

**Fig. 12****Fig. 13**

**Fig. 14****Fig. 15**

**Fig. 16****Fig. 17**